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Impact of Political Risk and Uncertainty on FDI in South Asia

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Abstract This study examines the potential effect of political risk and macroeconomic policy uncertainty on FDI in South Asia. To highlight the affect of political risk and macro policy uncertainty on FDI, we setup a theoretical framework based on oligopolistic and imperfect competition environment in host country. Autoregressive distributed lags (ARDL) is used to examine the impact of political risk and macro economic policy uncertainty index on FDI inflows. Macroeconomic policy uncertainty and political risk indices are constructed for this purpose. The long run results show negative affect of political risk and macroeconomic policy uncertainty indices on FDI inflows. Trade openness shows positive effect on FDI inflows only in short run while in long run it has negative impact on due lack of creditability regarding consistent trade liberalization policy and high trade cost. Furthermore, the market size significantly affects the inflow of FDI both in long run and short run which shows that FDI inflows in South Asia are mainly depend on market size. South Asian economies need to focus on political and macroeconomic factors along with FDI incentives policies to attract more FDI.

Keywords FDI · Trade openness · Political risk · Autoregressive distributed lag

JEL Classifications F21

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Introduction

In the era of globalization, countries are expanding their links in capital markets networks worldwide. The developing countries can make economic progress by capital accumulation through profitable investment projects by giving equal opportunities to local and foreign investors. The process of globalization enhances the importance of the FDI for developing countries. The FDI is considered as an engine for economic development, employment and productivity improvements through technological and managerial spillover in developing countries (Lipsey 2002). The objectives of economic development can be realized without opening the door for foreign investors. Therefore, the attraction of foreign capital is main priority for policy makers. However, the developing countries could not attract FDI inflows at the level achieved by transitional countries.

The literature highlighted various economic and noneconomic factors that determine the inflow of FDI (Gunseli et al. 2004; Martin and Wachtel 1993). Market size, population, capital formation, infrastructure facilities and productivity of factor of production and trade openness etc. are the main economic factors that influence the behavior of investors, hence determine the amount of FDI inflows.

Recent literature emphasis the importance of political uncertainty factors in attracting FDI. Political uncertainty or political risk is generated by number of factors like institutional quality, internal and external conflict, military and religion involvements in national politics, socioeconomic and investment conditions of host country in determining the inflows of FDI. Political risk can be accessed by considering all these factors in host country environment (Jun and Singh 1996). Busse and Hefeker (2005) have identified the importance of political risk as precondition for FDI inflows in host country by analyzing data on 83 developing countries. Thomas and Worrall (1994) have pointed that institutional risk can become a serious threat for investment contract enforcement signed between multinational corporations (MNCs) and host country government. In this way, the return on capital can be deteriorated in those countries having more chances of political risk. Thus; instead of focusing on traditional economic determinants, it become important to equally consider the political factors for FDI inflows.

Another strand of literature focuses on macroeconomic policy uncertainty in developing countries that may limit FDI inflows from developed countries. The macroeconomic policy uncertainty may include the uncertainty regarding monetary policy, fiscal policy and exchange rate policy. The literature highlights various channels through which macroeconomic policy uncertainty may affect FDI inflows. It is generally argued that higher inflation, indicator of macroeconomic policy uncertainty, makes it difficult for MNCs to predict prices in host country (Fisher 1993). A persistent increase in fiscal deficit in host country is another important channel that may increase uncertainty about fiscal policy and results inform of high taxation rates. A high taxation rates results in high input cost of capital and thus reduce the profit margin for MNCs and finally this reduction adversely impact FDI inflows in host countries.

In developing countries, FDI flows have raised after 1980, tremendously. FDI flows have increased from US $10,100 millions in 1986 to US $87,124 millions in

1994. In 2007, FDI flows in developing countries were around US $500 billion. FDI flows have also improved in South Asia on the average US $968.27 millions in 1989–1994 to US $49,176.6 millions in 2008 (UNCTAD 2010). South Asian countries like Bangladesh, India, Pakistan and Sri Lanka have introduced a wide range of incentives for local and foreign investors. These countries provide facility to foreign investors in avoidance of double taxation, foreign investment protection under the rule and regulation and removing restrictions for foreign investment in different sectors of economy. The share of FDI in South Asia as compare to other regions of the world remains low (Ghani et al. 2010). Now, question is why FDI inflows in South Asia remain low as compare to other region of the world. This is the question that required theoretical as well empirical investigation for South Asian Economies.

Ghani, et al. (2010) pointed out that the major bottlenecks faced by South Asia economies in attracting FDI are low institutional quality, bad infrastructure and high cost of doing business etc. Law and order situation has generally deteriorated in recent years that ultimately increase the transaction cost for business. Similarly, regulatory quality is very poor that is challenging issues in South Asia. The poor regulatory quality also increases the transparency problems and processing problem sin business operation. In addition to, macroeconomic policy uncertainty including increase in the inflation rate, persistent budget deficit and exchange rate fluctuations increase the cost of doing business that counter the gains from FDI in terms of volume in South Asia. Theses considerations raise the importance to analyze the role of political and macroeconomic policy determinants for FDI inflows in South Asia.

Based on this background, the prime objective of this study is to analyze the role of political risk as well as macroeconomic policy uncertainty in determining the FDI inflows in South Asia. To achieve this objective, we have assumed oligopolistic and imperfect competition environment in host country for theoretical model formulation.[1] More specifically, we also identify empirical links among political risk, macroeconomic policy uncertainty and FDI inflows.

Review of Literature

Busse and Hefeker (2005) investigated the effect of political risk and institutions on FDI for developing countries. They focused on panel data of 83 developing countries for the period of 1984–2003. They accounted all twelve political risk factors calculated by political risk service group from 1984 that not only cover political risk factor but also institutional factors. They analyzed the dynamic panel data by using fixed effect model and the problem of autocorrelation have also been addressed by GMM estimators. The results revealed that political risk factors included government stability, internal and external conflict, ethnic tensions, law and order situations and democratic rights played a very significant role in FDI

[1] The imperfect competition assumption have been reported for MNC's in following studies (Brander and Spencer 1984; Janeba 1996; Del 2009).

inflows for developing countries. They also checked the robustness of estimates but the significance of above mention variables remain the same.

Rios-Morales et al. (2009) assessed political risk is an important determinant for FDI. They developed theoretical link between good governance indicator and political risk. In this study good governance score was used as a proxy to estimate the political instability. Furthermore, the political instability among countries could be differentiated on the basis of more prominent factors including rule of law, control of corruption among all other factors. They found that political instability and voice and accountability effect significantly on FDI.

Rajan and Hattari (2009) explored the main determinants of FDI inflows to Asian countries concentrating on panel data over the period of 1990–2005 for 12 Asian emerging countries. They used gravity model approach to access the effect of political, location and other economic determinants of FDI. The data on political risk was taken form International Country Risk Guide (ICRG) source. They found that intra Asian trade, transportation cost and information asymmetries matters for Intra Asian FDI flows but more FDI outflows emerged from Asian countries to outside the world. The political risk, corporate taxes in Asian countries revealed as an important determinant for attraction FDI inflows in Asian countries.

Chan and Gemayel (2004) provided analytical analysis of various components of political risk for Middle East and North African (MENA) countries. This paper analyzed the relationship of political risk and FDI inflows over the period of 1991–1999. A fixed effect and random effect specification was used. The investment risks as well as political risk was found as an important determinant for FDI. The government should adopt preemptive measures that bring long term policy stability which ultimately reduce political risk and investment risk in MENA countries.

Kim (2006) investigated the relationship between FDI and political risk in Asian countries for the period of 1984–2002. A fixed effect model and random effect model was estimated to examine the important determinants for FDI. The study explored that bilateral investment treaties (BITs) is one of the important way to attract FDI in Asian countries but the expropriation of political risk counter balance the benefits of(BITs) for Asian countries. So, the political risk factors could not be ignored for FDI attractiveness.

Del (2009) investigated the relationship between exchange rate risks, political instability and FDI outflows from United States to 53 countries for the period of 1982–2005. An interactive term between political risk and exchange rate was found important significant factors for FDI. Hermes and Lensink (2001) evaluated the effect of policy uncertainty in context of FDI outflows. The time span used for this study 1971–1991 for LDC countries. The result showed that policy volatility had significant effect on capital outflows. The result of this study suggested that policy uncertainty decreases the expected wealth that ultimately becomes a major cause of capital flight from local country. A government could play an important role for policy consistency that ultimately encouraged FDI from outside the world.

Barrell et al. (2003) investigated the relationship between FDI and exchange rate risk by applying GARCH model methodology. This technique captured the multivariate context and conditional mean and variance of exchange rate. This study investigated that exchange rate volatility is linked with market concentration or

monopoly power that discourage FDI in host country. Data set from the period of 1982–2000 was used for European Countries.

By summarizing the literature, it can be concluded that political risk and macroeconomic policy uncertainty negatively effect FDI inflows. So, it becomes compulsory to analyze the role of macroeconomic policy variables uncertainty and political risk along with economic factors for FDI inflows.

Theoretical Model

According to recent economic literature besides economic factors for FDI inflows political factors are also important. A country level tax reforms, formal rules or informal rules and investment restrictions are important component of political risk that effect financial ability of multinational firms (MNCs) but also restrict free capital mobility in host country (Huang and Sternquist 2007; Mauro 1998). Similarly, macroeconomic policy uncertainty affects MNCs through profit margin and increases the cost of credits for MNCs (Lown and Morgan 2006).

To determine the link between of political risk, macroeconomic policy uncertainty and FDI, we assumed partial equilibrium model for identical MNCs that enter in foreign (or host) country.[2] The model starts by assuming that MNCs (J) face oligopolistic type of environment with imperfect competition abroad (Head 2001; Brander and Spencer 1984). MNCs (J) produced final goods (Q) in foreign countries (F) has paid input cost (CF) for final goods (Q) produced.

Suppose that each MNC faces inverse demand function in home country (H) which can be written as:

$$P_H = \delta - \phi F \tag{1}$$

where P_H represents the market price of product in home country, δ and ϕ are parameters respectively, F is used for market clearing condition. The production quantity for each MNC can be represented by Q_H and $F = JQ_H$ represents market clearing condition. The inverse demand function for each MNC in home country can be written as:

$$P_H = \delta - \phi Q_H \tag{2}$$

where Q_H represents the quantity demanded for final goods in home country. In terms of home country the profit function for MNC's can be written as:

$$\Pi = P_H Q_H - \overline{C_H} Q_H \tag{3}$$

where $\overline{C_H}$ marginal cost which is equal to average variable cost by assuming the linear production function for firms. Instead of considering a single factor that may effect marginal cost of multinational firms (J), Del (2009) investigate that exchange rate uncertainty increase the cost for FDI in foreign country. He incorporated exchange rate volatility factor in profit margin represented by Eq. (3). However,

[2] Our theoretical model is based on assumptions in following studies related to MNCs are Schmalensee (1976), Del (2009), Lahiri and Ono (1998), Lahiri and Mesa (2006).

theoretical and empirical literature also highlight the importance of policy uncertainty variables that may increase the cost for MNC's in foreign country.

Keeping in mind the policy uncertainty, we incorporate macroeconomic policy uncertainty index related variable that increase the cost of FDI. The macroeconomic policy uncertainty index variable includes (1) monetary policy uncertainty (2) fiscal policy uncertainty and (3) exchange rate uncertainty. Holland (1986) concluded that inflation uncertainty effect firm's input cost though labor market channel. A high inflation uncertainty means that high wage contracts between firms and employees and become a major source of cost for the firm in foreign country. Aizenman and Marion (1993) identified the theoretical and empirical importance of fiscal and monetary policy uncertainty for investment. The fiscal policy uncertainty decreases the expected marginal profit through tax rate channel that increase the cost for firms. Moreover, fiscal policy uncertainty has negative effect on investment decision.

The production cost of MNCs (J) is also affected by macroeconomic policy uncertainty in foreign country that ultimately affect the profit of MNCs. We incorporate macroeconomic policy uncertainty index that is composite of inflation rate uncertainty, budget deficit uncertainty and exchange rate uncertainty. The macroeconomic policy uncertainty index is denoted by η which is function of inflation rate uncertainty, budget deficit uncertainty and exchange rate uncertainty. i.e.:

$$\eta = f(InfU, BdfU, ExU) \tag{4}$$

where $InfU$ = Inflation rate uncertainty

$BdfU$ = Budget deficit uncertainty

ExU = Exchange rate uncertainty.

The marginal cost function faced by MNCs can be described as comparative to home production cost:

$$\overline{C_H} = \hat{\eta} C_F \tag{5}$$

By incorporating, the value in Eq. (3):

$$\Pi = P_H Q_H - \hat{\eta} C_F Q_H \tag{6}$$

MNCs also face political risk [$\rho \in (0, 1)$] in foreign country that also affects the production cost of final goods. Therefore Eq. (7) can be written as:

$$\Pi = (\delta - \phi J Q_H) Q_H - \hat{\eta} C_F Q_H - \rho Q_H \tag{7}$$

If, we assume that macroeconomic policy uncertainty which is composite of exchange rate uncertainty, inflation rate uncertainty and budget deficit uncertainty is normal random variable then we can apply the certainty equivalence of profit.[3] Assume that MNCs face partial equilibrium. The certainty equivalence profit for firm can be expressed as:

$$\Pi_{CE} = (P_H - \rho - M_H C_F) Q_H \tag{8}$$

where M_H that represents uncertainty or standard deviation of policy uncertainty index.

[3] This assumption is supported in following studies for exchange rate uncertainty (Del 2009; Cushman 1985; Lahiri and Mesa 2006; Goldberg and Kolstad 1995), for inflation uncertainty (Bhar and Hamori 2004) and for budget deficit uncertainty (Lensink et al. 1999).

The optimal output derives from profit maximization condition can be expressed as:

$$Q_H^* = \frac{\delta - \rho - M_H C_F}{\phi[J+1]} \tag{9}$$

Assuming that MNCs (J) take investment in small open economy and the investment decision can earn profit for each MNC that is in the form of reservation profit Π_R and free entry condition is assumed ($\Pi_{CE} = \Pi_R$). An expression form for MNCs entering in foreign country is a function of different parameters that ultimately capture the pattern of FDI from MNCs (J) in host country or developing country can be written after some derivation[4]

$$[J+1] = \frac{\delta - \rho - M_H C_F}{\sqrt{\phi \Pi_R}} \tag{10}$$

where reservation profit Π_R. The total differential of Eq. (10) captures the effect of macroeconomic policy uncertainty and political risk in foreign country or developing country on (J) multinational firms

Taking square on both side of Eq. (10).We get:

$$[J+1]^2 = \left(\frac{\delta - \rho - M_H C_F}{\sqrt{\phi \Pi_R}}\right)^2 \tag{11}$$

$$[J+1]^2 = \frac{(\delta - \rho - M_H C_F)^2}{\phi \Pi_R} \tag{12}$$

The total differential of Eq. (12) is given as:

$$2[J+1]d_J = -\frac{2(\delta - \rho - M_H C_F)}{\varphi \Pi_R} \cdot d\rho - \frac{2(\delta - \rho - M_H C_F)}{\phi \Pi_R} \cdot dM_H \tag{13}$$

The effect of change in policy uncertainty and its on FDI can be represented if we keep political risk parameter constant that is:

$$\frac{d_J}{d\sigma_\eta} = -\frac{(\delta - \rho - M_H C_F)}{\phi \Pi_R (J+1)} \frac{dM_H}{d\sigma_\eta} \angle 0 \tag{14}$$

From above expression it is clear that macroeconomic policy uncertainty exerts negative effect on FDI that may discourage FDI inflows to developing countries. Similarly, the effect of change in political risk in host country on FDI inflows can be represented by keeping macroeconomic policy uncertainty constant:

$$\frac{d_J}{d\rho} = -\frac{(\delta - \rho - M_H C_F)}{\phi \Pi_R (J+1)} \angle 0 \tag{15}$$

It is clear from Eq. (15), the political risk factor of host country effect negatively to the entrance of FDI inflows from (J) multinational firms. Based on above discussion, we conclude that the effect of political risk and macroeconomic policy uncertainty on FDI inflows from (J) multinational firms is negative.

[4] Detailed derivation of model can be seen in "Appendix".

Based on this model, we develop the empirical model by incorporating the economic factors, political risk, and macroeconomic policy uncertainty factor. According to Kamely (2004), there is no unanimous ideology accepted theoretically for FDI determinants'. An economic literature has given importance to economic variables for FDI attractiveness. Sahoo (2006) highlighted the role of economic factors for FDI inflows. He identified that market size; trade openness, infrastructure index and labor force growth rate as major determinants of FDI in South Asia. The market size is an important variable that attracts FDI inflows. Gross domestic product is an important proxy for market size of host country and it has positive effect on FDI inflows (Chakrabarti 2001; Mosa and Cardak 2006). The infrastructure of host country has its own empirical importance for FDI inflows. The energy consumption per capita has been used as a proxy for infrastructure that may have positive sign (Del 2009). Similarly, trade openness shows an ambiguous kind of relationship that may be positive or negative (Rahman 2003; Khan and Samad 2010; Aizenman and Noy 2005; Asiedu 2002). For empirical purpose, we specify the following model:

$$FDI_{it} = f(GDP_{it}, EPC_{it}, TO_{it}) \tag{16}$$

where

FDI = Foreign Direct Investment Inflows (at current US $)
GDP = GDP (at current us $).
EPC = Energy consumption per capita.
TO = Trade openness.

A political risk factor is composite of institutional factors as well as financial risk factors also. Political risk has expected negative sign. We have extended Eq. (16) by incorporating political risk index i.e.

$$FDI_{it} = f(GDP_{it}, EPC_{it}, TO_{it}, PRI_{it}) \tag{17}$$

where PRI = Political Risk Index.

Macroeconomic policy uncertainty variable is considered another important element for FDI inflows in host country. This macroeconomic policy uncertainty variable is composite of uncertainties regarding the monetary policy, fiscal policy and exchange rate policies. Hermes and Lensink (2001) evaluated the effect of policy uncertainty related to Inflation, exchange rate and budget deficit for lower developing countries (LDCs). We develop macroeconomic policy uncertainty index (PUI) that effect negatively to FDI inflows. Keeping this in view, macroeconomic policy uncertainty index can be incorporated in previous equation and final equation can be written as:

$$FDI_{it} = f(GDP_{it}, EPC_{it}, TO_{it}, PRI_{it}, PUI_{it}) \tag{18}$$

where PUI = Macroeconomic policy uncertainty Index.

Data and Econometric Methodology

Data

This study is based on panel data of four selected South Asian countries including, Bangladesh, India, Pakistan and Sri Lanka over the period of 1990–2008. The GDP

at current $ price was used as proxy for market size. We used trade openness (*TO*) for trade liberalization policy. We used electric power consumption per capita (*EPC*) as an indicator for infrastructure development in host country. The data on following economic variables (*GDP, EPC, TO*) is obtained from World Development Indicators (WDI). We used FDI inflows as percentage of GDP (*FDI*) as dependant variable. The data on (FDI) were taken from United Nation conference on Trade and Development (*UNCTAD*). The data on Inflation, exchange rate was taken from World development indicators (WDI) while the data on budget deficit as percentage of GDP was calculated from central banks reports of relevant countries. The data on political risk were taken form International Country Risk Guide (ICRG) source.

Econometric Methodology

Panel Unit Root

The augmented Dickey-Fuller (ADF) test has received considerable attention in time series data with non stationary null hypothesis. Levin et al. (2002) developed panel unit root test by extending called (LLC) panel unit root test. They incorporate heterogeneity of each cross sectional unit (N) deterministic effect and error term serial correlation structure with homogenous autoregressive assumption. LLC panel unit root have short comings that include independent assumption of each cross sectional unit that make LLC panel unit root test useless if cross correlation exists between cross sectional unit. Another limitation is auto regressive parameters are assumed same for cross sectional unit. Im et al. (1997, 2003) proposed a more advanced panel unit root test in heterogeneous panel data that depend upon reputed Dickey-Fuller procedure. This test is also applicable in case of non stationarity of cross section unit, heterogeneous serial correlation structure of error terms across each country unit and unbalanced panel data with some simulations is required. We used IPS test for panel unit root identification.

An ARDL Approach to Co Integration

The concept of co integration was developed by Engle and Granger (1987). Later on Johansen and Juselius (1990), Im et al. (1997), Pesaran et al. (2001) further developed the concept of co integration that was widely recognized.

This study estimates the long run relationship between FDI and its determinants represented by Eq. (18) utilizing ARDL approach developed by (Pesaran et al). An ARDL approach to co integration is more appropriate for simultaneous estimation of long run and short run relationship in a small finite sample that contained 30–80 values. An ARDL approach evaluates the existence of long run relationship among variables irrespective the order of integration of variables that may be zero order ($I(0)$), order one ($I(1)$) or mixed order form. This approach is applicable incase of endogenity and serial correlation problem and parameters estimated by this approach are unbiased and efficient (Pesaran et al. 2001). To estimate the long run

relationship, the autoregressive distributed lag ARDL model can be specified as model:

$$\Delta \ln FDI_{it} = \alpha + \sum_{k=1}^{p} \theta_i \Delta \ln FDI_{i,t-i} + \sum_{i=0}^{p} \varphi_i \Delta PRI_{i,t-i} + \sum_{i=0}^{p} \delta_i \Delta PUI_{i,t-i} + \sum_{i=0}^{p} \gamma_i \Delta EPC_{i,t-i} + \sum_{i=0}^{p} \omega_i \Delta TO_{i,t-i} + \sum_{i=0}^{p} \rho_i \Delta \ln GDP_{i,t-i} + \beta_1 \ln FDI_{i,t-1} + \beta_2 PRI_{i,t-1} + \beta_3 PUI_{i,t-1} + \beta_4 EPC_{i,t-1} + \beta_5 TO_{I,T-1} + \beta_6 \ln GDP_{i,t-1} + \mu_{it}$$

where $(\theta_i, \phi_i, \delta_i, \gamma_i, \omega_i, \rho_i)$ are the short run parameters, (β_1, β_2, β_3, β_4, $\beta_5\beta_6$) are the long run parameters.

The null and alternate hypothesis of no co integration can be formulated as:

H_0: $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = 0$)(No co integration)

Ha: $\beta_1 \neq \beta_2 \neq \beta_3 \neq \beta_4 \neq \beta_5 \neq \beta_6 \neq 0$)(Co integration).

Two sets of critical values are I(0) and I(1) are calculated by Pesaran et al. (2001). If calculated value is greater the upper bound critical value then null hypothesis of no cointegration can be rejected and the null hypothesis of no co integration can not be rejected if calculated F-statistical value lies below the lower critical value. Akaike Information Criteria (AIC) or Schwartz-Bayesian criteria are used to select the optimal lag length of each economic variable. When there is long run relationship among variables the short run dynamics are estimated through vector error correction model (VECM).The short run coefficients are estimated through error correction model. Error correction can be specified as:

$$\Delta \ln FDI_{it} = \alpha + \sum_{k=1}^{p} \theta_i \Delta \ln FDI_{i,t-i} + \sum_{i=0}^{p} \phi_i \Delta PRI_{i,t-i} + \sum_{i=0}^{p} \delta_i \Delta PUI_{i,t-i} + \sum_{ii=0}^{p} \gamma_i \Delta EPC_{i,t-i} + \sum_{i=0}^{p} \omega_i \Delta TO_{i,t-i} + \sum_{i=0}^{p} \rho_i \Delta \ln GDP_{i,t-i} + \beta EC_{i,t-1} + \mu_{it}.$$

In above mentioned Model, ($EC_{i,t-1}$) represent 1 year lagged period of error correction model. The coefficient of error correction model (β) measures the speed of convergence or divergence in case of shocks to or from long equilibrium. The significance of these coefficients confirms short run relationship and it is also an evidence for long run co integration of economic variables.

Empirical Results

Construction of Indices

We have estimated macroeconomic policy uncertainty index for all countries. In first step, uncertainties regarding inflation, exchange rate and budget deficit were calculated through variance approach. The variance of unpredicted part was

calculated from residual of univariate model (Aizenman and Marion 1993). The model can be written as:

$$Y_t = \alpha_o + y_{t-1} + y_{t-2} + trend + \varepsilon$$

By using this model, we estimated the uncertainty regarding fiscal policy, monetary policy uncertainty and exchange rate uncertainty respectively. Budget deficit uncertainty was used for proxy of fiscal policy uncertainty, inflation uncertainty was used for proxy of monetary policy and exchange rate uncertainty was used for exchange rate uncertainty. In second step, macroeconomic policy uncertainty index had been developed by principal component approach that captures the combined effect fiscal policy uncertainty, monetary policy uncertainty and exchange rate policy uncertainty into a single index form. We used inflation as CPI proxy for monetary policy, budget Deficit as a percentage of GDP for fiscal policy and average nominal exchange rate for calculation of macro economic policy uncertainty.

Political Risk of a country can be accounted by assigning numerical values of governance quality as well as sovereign risk evaluation of a country. We used political risk index that is composite of government stability, socioeconomic conditions, investment profile, internal conflict, external conflict, corruption, military in politics, religion in politics, law and order condition, ethnic tensions and democratic accountability of host country. Political risk index was generated by principal component approach (PCA). PCA basically capture the maximum variation in data with linear combination of set of variables by calculating eigen values and eigen vectors. Eigen vectors are arranged according to maximum variations towards lowest variations captured by eigen values in data.

Panel Unit Test Results

Before conducting cointegration analysis, it is important to check whether specified economic variables are stationary or non stationary. According to Asteriou and Hall (2007), if non stationary variables are used in econometric estimation that may lead to spurious regression results. To check the stationary of variables, we apply Im et al. (1997) test and w-statistics is used for this purpose. IPS test is performed on some conditions that may include (a) individual intercept but no trends (b) individual Intercept with trend and (c) one lag is assumed in all variables. The results of IPS test at level and first difference are reported in Table 1.

The results show that political risk index (PRI_{it}) and policy uncertainty index (PUI_{it}) are stationary by each specification of constant and constant and trend. This implies that the null hypothesis of unit root are rejected for PRI_{it} andPUI_{it} at level. Hence theses variables are integrated of order zero i.e. ($I(0)$). While the other variables including foreign direct investment inflows ($LFDI_{it}$), trade openness (TO_{it}), gross domestic product ($LGDP_{it}$) and electric power consumption per capita (EPC_{it}) are nonstationary at level and stationary at their first difference. Thus we conclude that$LFDI_{it}$, TO_{it}, $LGDP_{it}$ and EPC_{it} are integrated of order one i.e. $I(1)$.

Table 1 IPS panel unit root test of stationary

	Im et al. at Level		Im et al. (W-stat) at Ist difference	
	Without trend	With trend	Without trend	With trend
$LFDI_{it}$	1.1325 (0.8719)	−0.29608 (0.3836)	−6.83 (0.0000)*	−6.07 (0.0000)*
PRI_{it}	−1.3082 (0.0954)***	−2.4492 (0.0072)*	−2.7929 (0.0026)*	−1.8250 (0.034)**
PUI_{it}	−4.94 (0.000)*	−3.39 (0.0003)*	−8.389 (0.0000)*	−6.42 (0.000)*
TO_{it}	2.5721 (0.9949)	1.8044 (0.9644)	−5.549 (0.000)*	−5.100 (0.000)*
$LGDP_{it}$	6.34 (1.00)	0.8265 (0.7958)	−2.1559 (0.0156)*	−1.6029 (0.1230)
EPC_{it}	2.87502 (0.9980)	0.57140 (0.7161)	−1.80283 (0.0357)**	−1.05805 (0.1450)

Note: *, **, *** indicate the significance at 1, 5, 10% respectively, The value in parenthesis are the p-value

Table 2 Selection of lag length

Lag	LogL	LR	FPE	AIC	SC	HQ
0	−1,324.84	NA	1.68e + 08	35.96886	36.15567	36.04338
1	−1,023.34	545.9691	129,207.9	28.79304	30.10075*	29.31470*
2	−982.36	67.56367*	115,063.9*	28.65841*	31.08702	29.62721

* Indicates lag order selected by the criterion

LR sequential modified LR test statistic (each test at 5% level), *FPE* Final prediction error, *AIC* Akaike information criterion, *SC* Schwarz information criterion, *HQ* Hannan-Quinn information criterion

Due this mixed order of integration the most appropriate approach for estimation of long run relationship in our study is Auto regressive distributed lag (ARDL) approach to co integration.

Auto Regressive Distributed Lag (ARDL) Approach

Before applying Auto regressive distributed lag (ARDL) Approach, the first step is to select optimal lag length. We select two lags on the basis of Akaike information critera (AIC). The results are given in Table 2.

Based on AIC two lags are selected for vector auto regressive process of co integration. After the lag selection, we have estimated unrestricted error correction model specified in Eq. (18) through Panel Least Squares method. The results are reported in Table 3.

To find the co integration, in the next step we calculate F-statistics by applying wald- test. The computed F-statistics is compared with upper bound F-statistics critical value computed by Pesaran et al. (2001). The computed F-statistics value is 6.07 is higher than upper bound critical F-statistics value i.e. 3.99. The null hypothesis of no co integration is rejected and it implies long run relationship between $LFDI_{it}$, TO_{it}, $LGDP_{it}$, PRI_{it}, PUI_{it}. To obtained long run coefficients, we normalize the lag level variables by dividing on the coefficient of $LFDI_{it}$ and

Table 3 ARDL approach results (dependant variable: $\Delta LFDI_{it}$)

Variables	Coefficients	t-statistics
ΔPRI_{it}	−0.109**	−2.38
$\Delta PRI_{it\text{-}1}$	0.925**	2.22
$\Delta PRI_{it\text{-}2}$	0.083***	1.96
ΔPUI_{it}	−0.63**	−2.19
ΔEPC_{it}	0.0023**	2.11
ΔTO_{it}	0.035*	2.67
$\Delta TO_{it\text{-}1}$	0.040*	3.59
$\Delta TO_{it\text{-}2}$	0.027**	2.27
$\Delta LGDP_{it}$	1.31*	8.97
$LFDI_{it\text{-}1}$	−0.38*	−5.11
$PRI_{it\text{-}1}$	−0.072*	−3.24
$PUI_{it\text{-}1}$	−0.091*	−2.06
$EPC_{it\text{-}1}$	0.0021*	3.12
$TO_{it\text{-}1}$	−0.016**	−2.13
$LGDP_{it\text{-}1}$	0.24***	1.99
Constant	1.23	0.87
R^2	0.82	
Adjusted R^2	0.78	
F-Statistics	18.05 (0.000)	
Durbin Watson	2.02	
Normality test	0.660 (0.718)	

Note: *, **, *** indicate the statistical significance at 1, 5, 10% respectively

assumed all other variables equal to zero.[5] In this way, the long run elasticity's are obtained and normalization equation is reported below:

$$\ln FDI_{it} = -0.18PRI^{*}_{it} - 0.23PUI^{*}_{it} + 0.0055EPC^{*}_{it} - 0.041TO^{**}_{it} + 0.61 \ln GDP^{***}_{it}$$

The results show that political risk exerts negative and affect on FDI inflows in long run. This implies that weak institutional structure that includes unstable government, corruption, internal and external conflict religious and poor law and order situation which are unwanted charters tics of South Asian countries effect negatively to FDI inflows. These factors increase the transaction cost for foreign investors and increase incredibility of host country to protect the assets for MNCs.

Macroeconomic policy uncertainty coefficient value is −0.23 that effect negatively on FDI inflows in long run. It means that macro economic policy uncertainty is composite of budget deficit; exchange rate uncertainty and inflation uncertainty adversely effect FDI inflows in long run. The finding suggests that macroeconomic policy uncertainty is an important factor for FDI because it affect profit margin by increasing cost for foreign firms. Macroeconomic policy uncertainty effect MNCs by increasing cost of production through price channel and imposing of extra taxes in host country. Electric power consumption per capita

[5] A steady state condition is assumed for normalization. *, **, *** indicate the statistical significance at 1, 5, 10% respectively.

used as proxy for infrastructure effect positively and its coefficient is 0.0055 which is significant in long run. This factor indicates that as infrastructure facility in host country increase it attract more FDI from different parts of world. The long run coefficient of GDP is 0.61 and significant. It means the large market size show positive impact on FDI inflows. The increase in GDP generates the demand of goods and services in host country. The increase in demand of goods and services provide incentives for local and foreign investors to achieve economies of scale in production (Khan and Samad 2010). So, large market size attract more FDI.

The long run coefficient of trade openness coefficient is −0.041. The impact of trade openness on FDI inflows is significant though negative. The sign of trade openness in long run is consistent with common empirical literature. The negative effect of trade openness is also justified that a risk and uncertainty factor has also given theoretical importance in investment theory. This theory has provided a new paradigm that investor either to invest or gain through taking right decision of investment by accessing risk of investment. According to Rahman (2003), lack of credibility regarding consistent trade liberalization adopted by developing countries is one of the main reason that effect negatively to foreign investor decision to take long run investment decision. According to real option theory and investors' risk theory, foreign investors are more sensitive about risk and uncertainty factors about host country that depict credibility of that country regarding trade liberalization policies. So, foreign investors save the cost by taking decision not to invest in more risky countries comparative to domestic investors cost in host country (Lehmann 1999).

Another way to explain the effect of Trade openness on cost of production is through the wage differential channel. It is argued that trade liberalization policy increase the wage differential between skilled and unskilled labor. MNC's face domestic competition and that force MNC's to cut the cost of production. High wage differential increases the cost of production in host country. A high level cost of production discourages FDI inflows (Becker 1971).

The coefficient value of political risk that is (0.18) and macroeconomic policy uncertainty coefficient value is (0.23) is higher comparative to trade openness coefficient (0.041), electric power consumption value (0.0055). On the basis of coefficient values, it can be argued that political risk and macroeconomic policy uncertainty matters more for FDI. Similarly, the value of market size coefficient is higher in long run that is 0.61. This implies that most of FDI inflows in South Asia is on the basis of large market size.

Error Correction Model for Short Run Analysis

To examine the short run dynamics we estimate restricted error correction model. The short run results are reported in Table 4.

The error correction term depicts the convergence or divergence of series. The size of error correction term (ECit-1) is −0.26 which is negative and significant. It means that calculated t-value is above the critical value that implies the rejection of null hypothesis of no short run relationship exists. Thus we can imply that short run relationship exists that also confirms existence of long run relationship. In case of

Table 4 Error correction results. dependant variable: $\Delta LFDI_{it}$

Variables	Coefficients	t-statistics
ΔPRI_{it}	−0.04*	−2.64
ΔPUI_{it}	−0.63*	−3.05
ΔEPC_{it}	0.0022**	2.22
ΔTO_{it}	0.033*	2.72
$\Delta LGDP_{it}$	1.37*	10.12
ECit-1	−0.26*	−5.49
Constant	0.85*	5.32
R^2	0.78	
Adj R^2	0.76	
F-Stat	34.70 (0.000)	

Note: *, **, *** indicate the statistical significance at 1, 5, 10% respectively

shocks all variables will converge to their long run equilibrium path with 26 % annual speed of adjustment. The political risk shows negative and significant effect in short run. It means that as political risk increases in short run FDI inflows decreases in South Asia. The policy uncertainty coefficient is negative and significant. FDI inflows are more sensitive to macro economic policy uncertainty in short run as compared to long run. The impact of GDP on FDI inflows is positive in short run. The impact of trade openness on FDI inflows is positive in short run. Trade openness indicates that it generates demand for imports required capital goods and services in short run that attract more FDI inflows (Khan and Samad 2010). The sign of trade openness is generally consistent to economic literature in short run (Aizenman and Noy 2005; Asiedu 2002).

Conclusion and Policy Recommendation

The main objective to this is to assess the impact of political risk and macroeconomic policy uncertainty along with other economic factors on FDI in selected south Asian countries. An ARDL approach to co-integration technique is used for estimation of long run relationship of economic variable and for and for short run relationship Error correction model is applied.

Before conducting empirical analysis, political risk and policy uncertainty index have been developed for south Asia. These indices are used further to check relevant importance in context of FDI inflows. According to the results of the study, we find long run relationship between political risk index and FDI inflows. Political risk index also matters in short run for FDI inflows. The econometric results of this variable is according to our hypothesis of the study. The relationship between political risk and FDI inflows must be prioritized along with FDI incentives policies. Political risk includes government stability, corruption environment, internal and external conflict and democratic accountability etc. The results show that political risk have their own substantial importance for FDI inflows both in long run and short run for South Asia.

In addition to political risk, the importance of macro economic policy uncertainty has been confirmed for FDI inflows. Our study also is in favor of the view that macroeconomic policy uncertainty effect negatively on FDI inflows in selected South Asian countries. Macro economic policy uncertainty normally discourages the foreign investors and creates an unstable or uncertain situation about the host country environment. Macro economic policy uncertainty regarding budget deficit, inflation and exchange rate normally discourages the investor to bring investment in South Asia. The impact of trade openness on FDI inflows in negative in long run that maybe due to lack of creditability regarding trade liberalization policy and high trade cost in host country that discourage FDI inflows in long run while the impact of trade openness on FDI inflows is positive in short run. A large market size exerts positive impact on FDI inflows in long run and short run. An electric power consumption variable also shows positive importance in long and short run context. For attracting FDI from other part of the World, South Asian countries should also focus political risk factors along with macroeconomic policy uncertainty factors. A political risk environment and macro economic policy uncertainty should be given substantial importance in FDI policies.

Appendix

An Empirical work before Eq. (8) is based on assumption that MNCs face Certainty equivalence of profit and Policy uncertainty variables are normally distributed in empirical literature including the work of Aizenman and Marion (1993), Lahiri and Mesa (2006). The profit function can be described in case of certainty equivalence can be written as:

$$\Pi_{CE} = (P_H - C_H - \rho)Q_F - \psi StD(C_H Q_H)$$

The derivation before Eq. (9)

$$\Pi_{CE} = P_H Q_H - \rho Q_H - M_H C_F Q_H$$

where $P_H = \delta - \phi J Q_H$ for Jth firms

The profit maximization condition in case of cournot equilibrium environments are:

$$\Pi_{CE} = (\delta - \phi J Q_H)Q_H - \rho Q_H - M_H C_F Q_H$$

The profit maximization conditions in case of cournot conjuncture for firm's environments are:

$$\frac{\partial \Pi_{CE}}{\partial Q_H} = 0 \text{ and } \frac{\partial \Pi_{CE}}{\partial Q_H} = 0$$

The maximization equation in case of cornout conjuncture for firms with respect to QH:

$$\delta - \phi J Q_H - \phi Q_H - \rho = M_H C_F$$

$$\delta - (J+1)\phi Q_H - \rho = M_H C_F$$

and further we can write as

$$P_H - \phi Q_H - \rho = M_F C_H$$

$$P_F - \rho - M_H C_F = \phi Q_H$$

The derivation procedure before Eq. (10). The certainty equivalence profit has been written in following form

$$\Pi_{CE} = (P_H - \rho - M_H C_F) Q_H$$

$$P_F - \rho - M_H C_F = \phi Q_H$$

so,

$$\Pi_{CE} = (\phi Q_H) Q_H = \phi Q_H^2$$

According to free entry condition

$$(\Pi_{CE} = \Pi_R)$$

It can be written as

$$(\Pi_{CE} = \Pi_R = \phi Q_H^2)$$

Putting the value of optimal value of output

$$\Pi_{CE} = \phi \left(\frac{\delta - \rho - M_H C_F}{\phi (J+1)} \right)^2$$

$$\Pi_R = \phi \left(\frac{\delta - \rho - M_H C_F}{\phi (J+1)} \right)^2$$

$$\Pi_R = \phi \left[\frac{(\delta - \rho - M_H C_F)^2}{(\phi (J+1)^2} \right]$$

$$\Pi_R = \frac{(\delta - \rho - M_H C_F)^2}{\phi (J+1)^2}$$

Taking cross multiplication on both sides of Equation

$$(J+1)^2 = \frac{(\delta - \rho - M_H C_F)^2}{\phi \Pi_R}$$

Taking square root on both sides of Equation

$$\sqrt{(J+1)^2} = \sqrt{\frac{(\delta - \rho - M_H C_F)^2}{\phi \Pi_R}}$$

$$[J+1] = \frac{\delta - \rho - M_H C_F}{\sqrt{\phi \Pi_R}} \tag{19}$$

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